

若 $f(x)$ 周期为 $2l$ 周期变化

设 $h(x) = f(2l \cdot \frac{x}{2\pi})$ 周期为 $2\pi = f(\frac{1}{l} \cdot x)$

$$h(x+2k\pi) = f(2l \cdot \frac{x+2k\pi}{2\pi}) = f(2l \cdot \frac{x}{2\pi}) = h(x)$$

$$f(x) = h(\frac{\pi}{l}x)$$

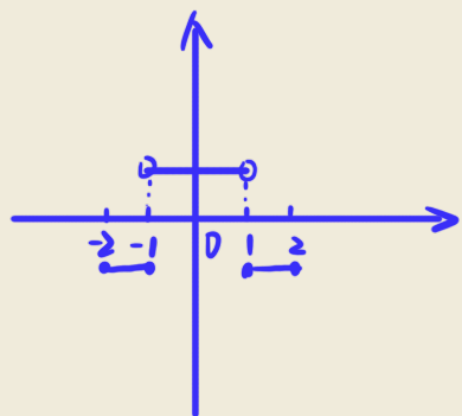
$$h(x) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(\frac{n\pi}{l}x) + \sum_{n=1}^{\infty} b_n \sin(\frac{n\pi}{l}x)$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} h(x) \cos nx \, dx, \text{ 设 } u = \frac{1}{l}x, x = \frac{\pi}{l}u$$

$$= \frac{1}{\pi} \int_{-l}^l h(u) \cos(\frac{n\pi}{l}x) \, dx$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin(\frac{n\pi}{l}x) \, dx$$

设 $f(x)$ 为周期为 4 的周期函数, $f(x) = \begin{cases} -1 & x \in [-2, 1] \\ 1 & x \in (-1, 1) \\ -1 & x \in [1, 2] \end{cases}$ 周期 $2l$



$$l=2, a_0 = \frac{1}{2} \cdot 2 \left(\int_0^1 dx + \int_1^2 (-1) dx \right) = 0$$

$$a_n = \frac{1}{2} \cdot 2 \left(\int_0^1 \cos(\frac{n\pi}{l}x) dx + \int_1^2 -\cos(\frac{n\pi}{l}x) dx \right)$$

$$= \frac{2}{n\pi} \sin(\frac{n\pi}{2}) \Big|_0^1 - \frac{2}{n\pi} \sin(\frac{n\pi}{2}) \Big|_1^2 = \frac{4}{n\pi} \sin(\frac{n\pi}{2})$$

$$= \begin{cases} 0 & n=2k \\ \frac{(-1)^k}{(2k+1)\pi} & n=2k+1 \end{cases}$$

$$f(x) \sim \sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{(2k-1)\pi} \cos\left(\frac{(2k-1)\pi}{2}x\right)$$

$$f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)\pi} \cdot \cos\left(\frac{(2n-1)\pi}{2}x\right), \quad x \in [-2, 1) \cup (-1, 1) \cup (1, 2]$$

将 $f(x) = x - [x]$ 展开成傅里叶级数

$f(x)$ 是周期为 1 的函数 $\Rightarrow l = \frac{1}{2}$ 除去 $x = k$ ($k \in \mathbb{Z}$) 之外, $f(x)$ 连续

$$a_0 = \frac{1}{\frac{1}{2}} \int_0^1 f(x) dx = 2 \int_0^1 (x - [x]) dx = 2 \int_0^1 x dx = 1$$

$$a_n = 2 \int_0^1 x \cdot \cos 2n\pi x \cdot dx = 2 \left(\frac{x}{2n\pi} \sin 2n\pi x + \frac{1}{4(n\pi)^2} \cos 2n\pi x \right) \Big|_0^1 = 0$$

$$b_n = 2 \int_0^1 x \sin 2n\pi x \cdot dx = 2 \left(-\frac{x}{2n\pi} \cos 2n\pi x + \frac{1}{4(n\pi)^2} \cdot \sin 2n\pi x \right) \Big|_0^1 = -\frac{1}{n\pi}$$

$$\frac{1}{2} - \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{\sin 2n\pi x}{n} = \begin{cases} x - [x] & x \neq k \\ \frac{1}{2} & x = k \end{cases} \quad k \in \mathbb{Z}$$